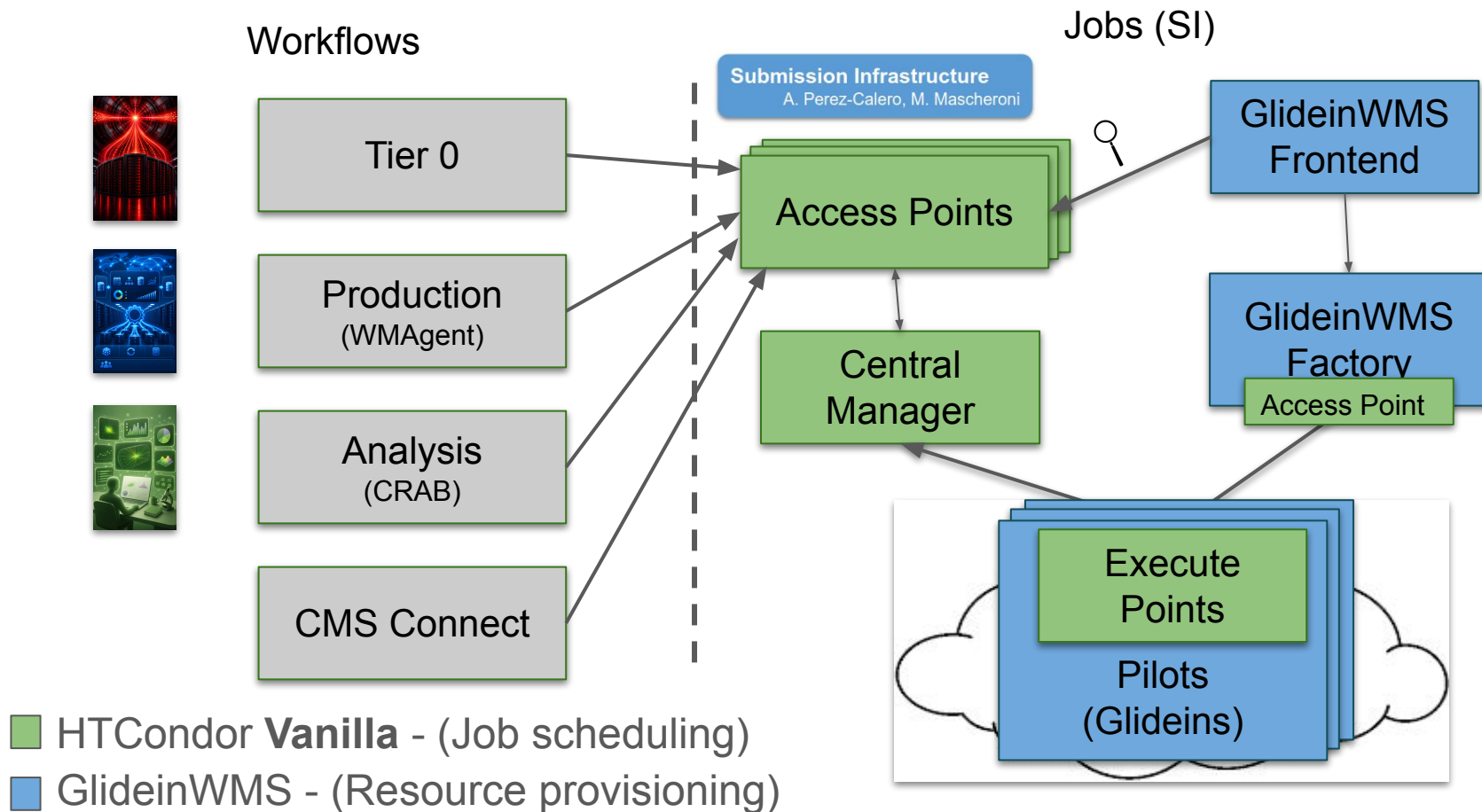


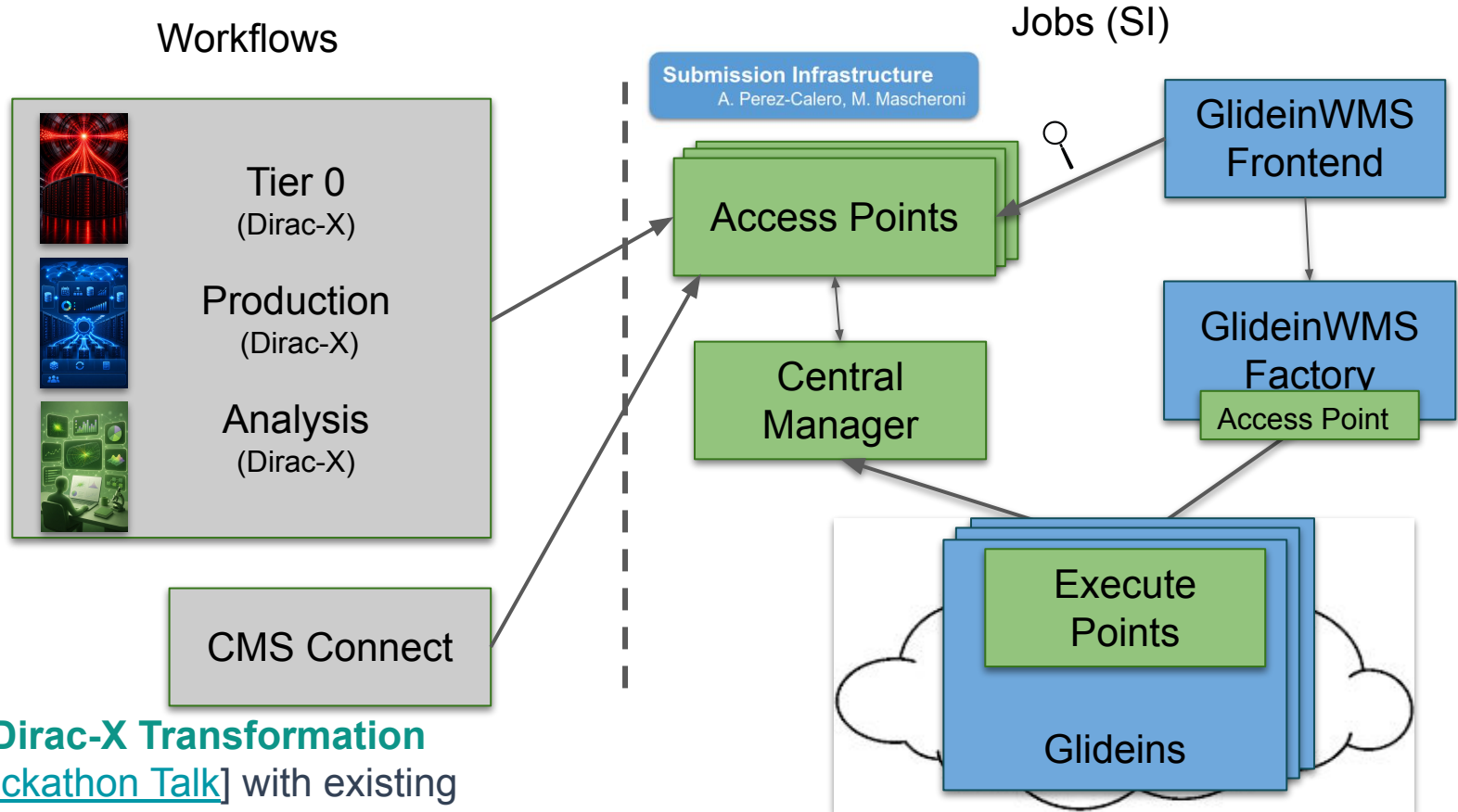
Exploring Collaboration Between HTCondor and DiracX

Marco Mascheroni

Current Architecture

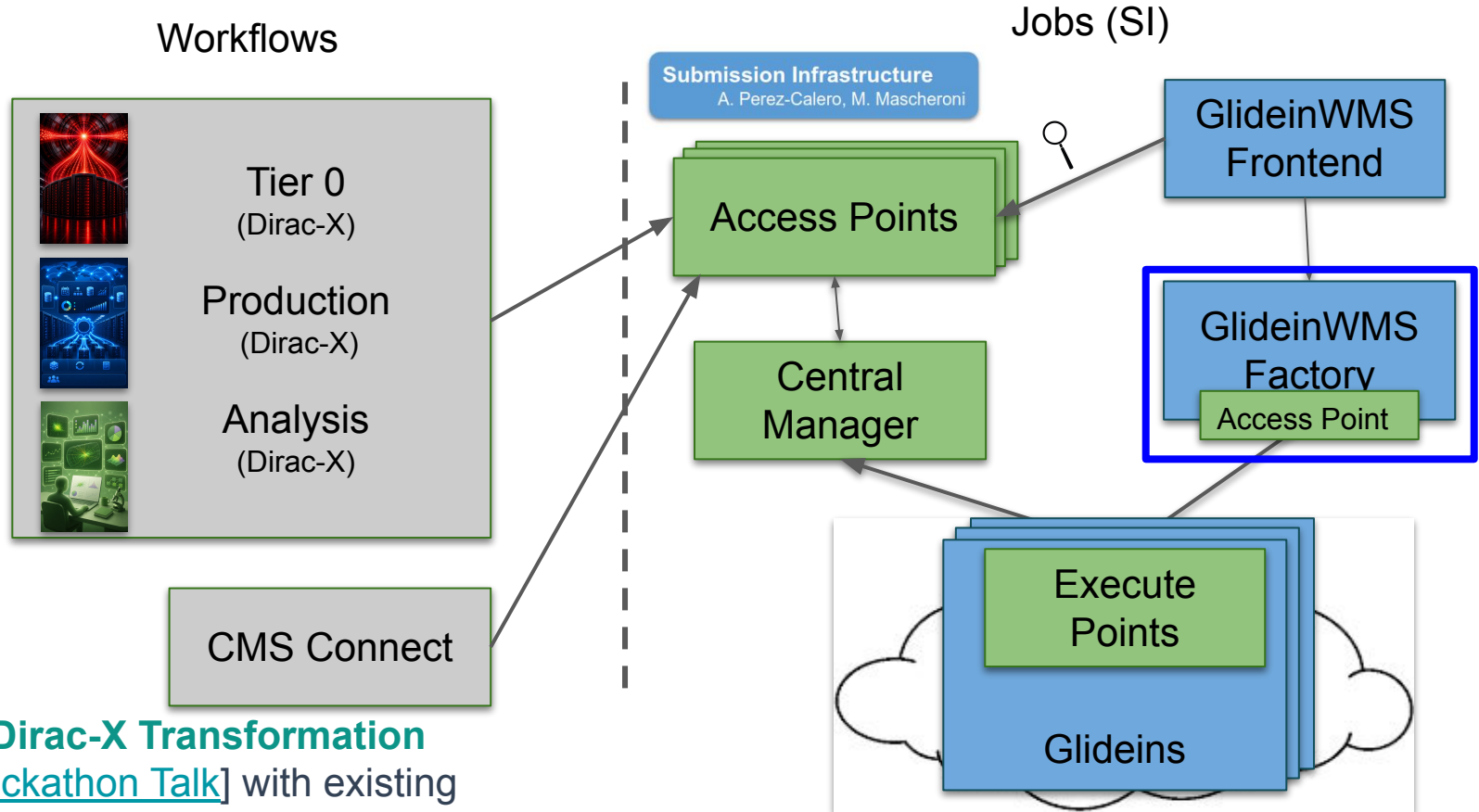


Direction for HL-LHC



Integrating **Dirac-X Transformation system** [[Hackathon Talk](#)] with existing glideinWMS & HTCondor infrastructure.

Direction for HL-LHC



Integrating **Dirac-X Transformation system** [[Hackathon Talk](#)] with existing glideinWMS & HTCondor infrastructure.

GlideinWMS Factory

- VO agnostic component responsible of submitting(managing) pilot jobs to CEs
- Receives credentials from VO specific GlideinWMS frontends
- Submits pilots to a local HTCondor scheduler
 - HTCondor-CEs
 - ARC-CEs
 - Slurm (via SSH)

```
[mmascher@vocms0206 ~]$ condor_q -name schedd_glideins3@vocms0206.cern.ch -all -limit 10 2> /dev/null
```

```
-- Schedd: schedd_glideins3@vocms0206.cern.ch : <188.185.22.37:9615?... @ 07/14/26 15:06:14
```

ID	FRONTEND	OWNER	SUBMITTED	RUN TIME	ST	PRI	SIZE	CMD
2539764.0	fecmsglobal		7/13 20:51	0+11:23:16	R	0	9766	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_UK_London_IC_ceprod01 -clientnam
2539764.1	fecmsglobal		7/13 20:51	0+11:22:52	R	0	9766	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_UK_London_IC_ceprod01 -clientnam
2540144.0	fecmsglobal		7/14 00:24	0+00:00:00	I	0	0.1	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_IT_Pisa_ce0 -clientname CMSG-v1
2540799.0	fecmsglobal		7/14 06:23	0+08:40:56	R	0	0.1	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_RU_JINR_lcgce01 -clientname CMSG
2540799.3	fecmsglobal		7/14 06:23	0+08:40:56	R	0	0.1	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_RU_JINR_lcgce01 -clientname CMSG
2540799.4	fecmsglobal		7/14 06:23	0+08:40:56	R	0	0.1	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_RU_JINR_lcgce01 -clientname CMSG
2540799.5	fecmsglobal		7/14 06:23	0+00:00:00	H	0	0.1	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_RU_JINR_lcgce01 -clientname CMSG
2540799.7	fecmsglobal		7/14 06:23	0+08:39:47	R	0	0.1	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_RU_JINR_lcgce01 -clientname CMSG
2541745.0	fecmscernt0		7/14 14:44	0+00:00:00	I	0	0.1	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_CH_CERN_ce505 -clientname CMS_T0
2541745.1	fecmscernt0		7/14 14:44	0+00:00:00	I	0	0.1	glidein_startup.sh -v std -name gfactory_instance -entry CMSHTPC_T2_CH_CERN_ce505 -clientname CMS_T0

InterCEde

A common abstraction for interacting with Computing Elements and batch systems

- Single Python interface
- Multiple backend implementations (ARC, HTCondor-CE, Slurm, ...)
- Backend-independent transformations
- Integration-tested backends

From Github repository:

<https://github.com/DIRACGrid/intercede>

What is interCEde?

interCEde sits between Workload Management Systems, such as [DiracX](#) and the many resources where jobs actually run. It provides a single, consistent interface for talking to **Computing Elements (CEs)** and **batch systems** — submitting jobs, querying their status, and retrieving their outputs — regardless of which backend is on the other end.

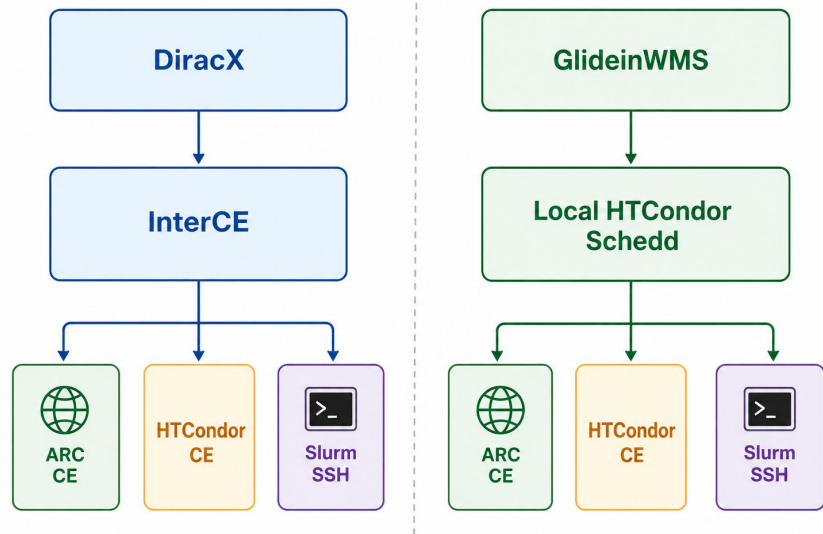
The name is the job description: the library *intercedes* on WMS' behalf, acting between two parties so the rest of the stack never has to know whether it is talking to ARC, HTCondor, Slurm over SSH, or a process on the local machine.

Every interface ships with **integration tests that run against containerized instances** of the real backends, so a given CE type is verified against multiple versions and configurations rather than against a mock that drifts from reality.

Similarities between InterCEde and HTCondor-G

Capability	InterCE	HTCondor
Submit jobs	✓	✓
Query job status	✓	✓
Cancel jobs	✓	✓
Retrieve job output	✓	✓
Support ARC CE	✓	✓
Support HTCondor-CE	✓	✓
Support Slurm	✓	✓
Extensible to additional resource types	✓	✓

InterCEde and HTCondor appear to provide a very similar resource submission abstraction.



Discussion: *Could HTCondor serve as the backend implementation of the InterCE abstraction?*

Summary

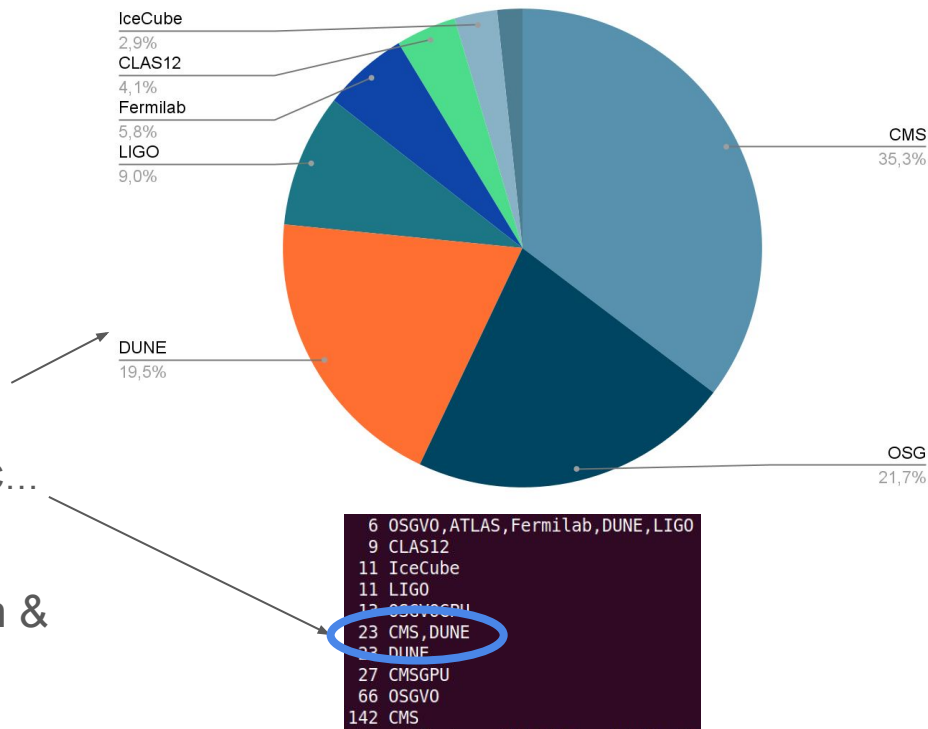
- Similarities in InterCEde and GlideinWMS Factories
 - Both need to submit pilot jobs to CEs
- InterCEde current direction appears to be implementing resource-specific backends for arsub, condor_ce_submit, ssh+sbatch to talk to CEs and HPCs batch systems
- GlideinWMS factory relies on condor_submit to a local scheduler (which in turn talks to CEs and HPCs)
- Could WLCG benefit from a shared submission layer?
 - ATLAS already uses condor (condor_submit to local schedulers)
 - If a critical mass forms maybe we can ask CERN IT to run a shared cross experiment component to submit pilots?
 - WLCG Experiments Test Framework (ETF) already check submissions across all T1s and T2 for CMS (and ATLAS?) using HTCondor-G

Additional Material

Shared Factories

CMS uses both dedicated and shared glideinWMS factories, taking advantage of **collaboration with other experiments**.

- 4 GlideinWMS factories
 - **CERN** and **FNAL**: CMS Only
 - **Tiger**: CMS + OSG + Other experiments
 - **Tempest**: OSG + Other experiments
- ~500 active entries in the OSG factory
 - 208 CMS, 128 OSG, 115 DUNE, 53 LIGO
 - 34 Fermilab, 24 CLAS12, 17 IceCube, 10 EIC...
 - Many entries are shared across experiments
- Operated by 2 FTEs across
- Advantage: **shared software validation & testing**



Sample JDL file

```
[mmascher@vocms0206 ~]$ cat /data/lib/gwms-factory/work-dir/entry_CMSHTPC_T2_CH_CERN_ce504/job.condor | grep -v Glidein
# File: job.condor
#
+SciTokensFile = "$ENV(SCITOKENS_FILE)"
environment = "JOB_TOKENS='/data/lib/gwms-factory/server-credentials/entry_CMSHTPC_T2_CH_CERN_ce504/tokens.tgz'"
Universe = grid
Grid_Resource = condor ce504.cern.ch ce504.cern.ch:9619
Executable = glidein_startup.sh
copy_to_spool = True
Arguments = $ENV(GLIDEIN_ARGUMENTS)
transfer_input_files = $ENV(IDENTITY_CREDENTIALS:),/data/lib/gwms-factory/server-credentials/entry_CMSHTPC_T2_CH_CERN_ce504/tokens.tgz
encrypt_input_files = $ENV(IDENTITY_CREDENTIALS:),/data/lib/gwms-factory/server-credentials/entry_CMSHTPC_T2_CH_CERN_ce504/tokens.tgz
Transfer_Executable = True
transfer_output_files =
WhenToTransferOutput = ON_EXIT
stream_output = False
stream_error = False
+TransferOutput = ""
x509userproxy = $ENV(X509_USER_PROXY:)
+maxMemory = 32000
+xcount = 16
+fename = "$ENV(GLIDEIN_USER)"
Notification = Never
+Owner = undefined
Log = /var/log/gwms-factory/client/user_${ENV(GLIDEIN_USER)}/glidein_gfactory_instance/entry_CMSHTPC_T2_CH_CERN_ce504/condor_activity_${ENV(GLIDEIN_LOGNR)}_${ENV(GLIDEIN_CLIENT)}.log
Output = /var/log/gwms-factory/client/user_${ENV(GLIDEIN_USER)}/glidein_gfactory_instance/entry_CMSHTPC_T2_CH_CERN_ce504/job.$(Cluster).$(Process).out
Error = /var/log/gwms-factory/client/user_${ENV(GLIDEIN_USER)}/glidein_gfactory_instance/entry_CMSHTPC_T2_CH_CERN_ce504/job.$(Cluster).$(Process).err
Queue $ENV(GLIDEIN_COUNT)
```